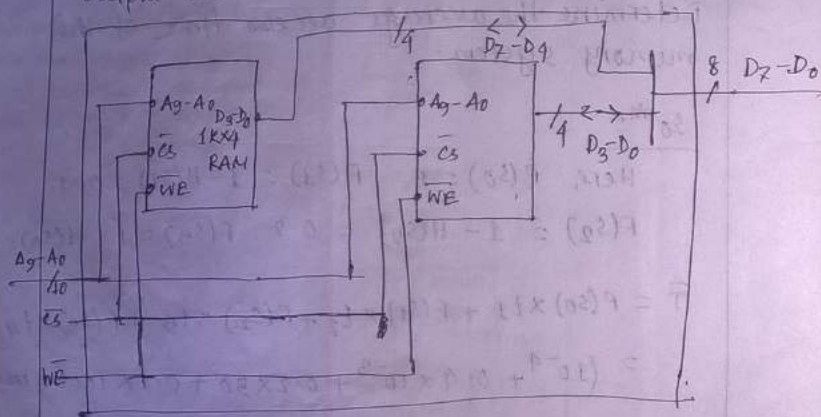


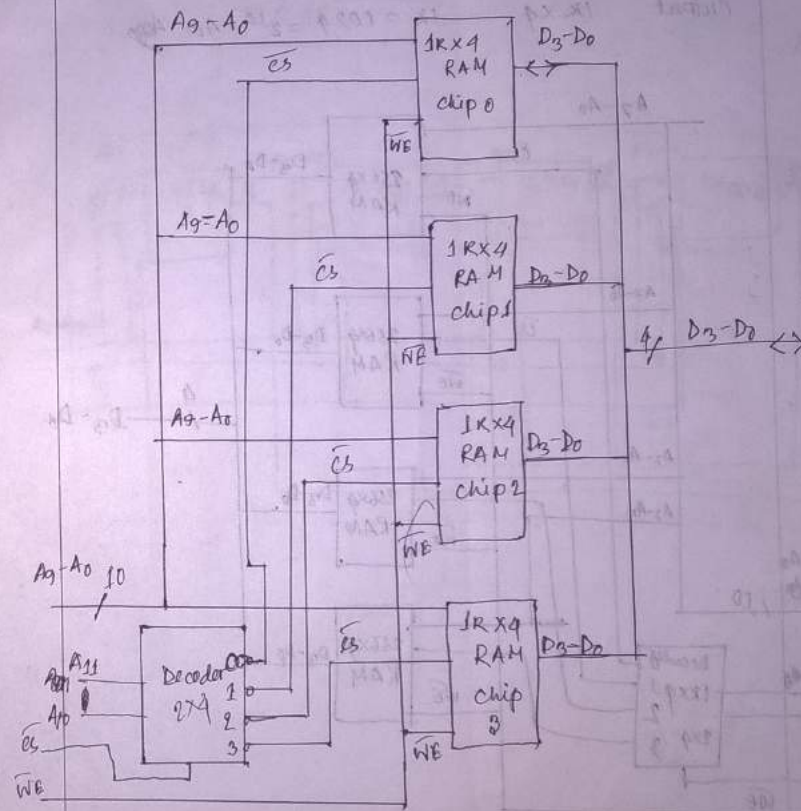
Memory Design:

** Design a $1K \times 8$ RAM using two $1K \times 4$ RAM chips.

Input $1K \times 4$
output $1K \times 8$.



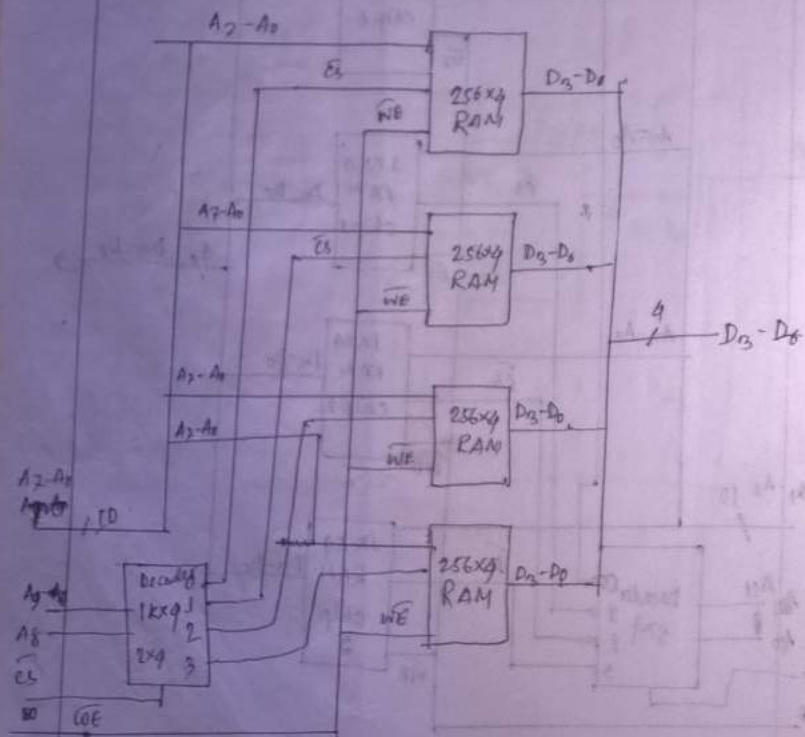
* Design $4K \times 4$ RAM using 4 $1K \times 4$ RAM chips.



* Design $1K \times 9$ using four 256×9 $1K = 1024$
 2^{10}

Input 256×9 $256 = 2^8$ $A_0 - A_7$

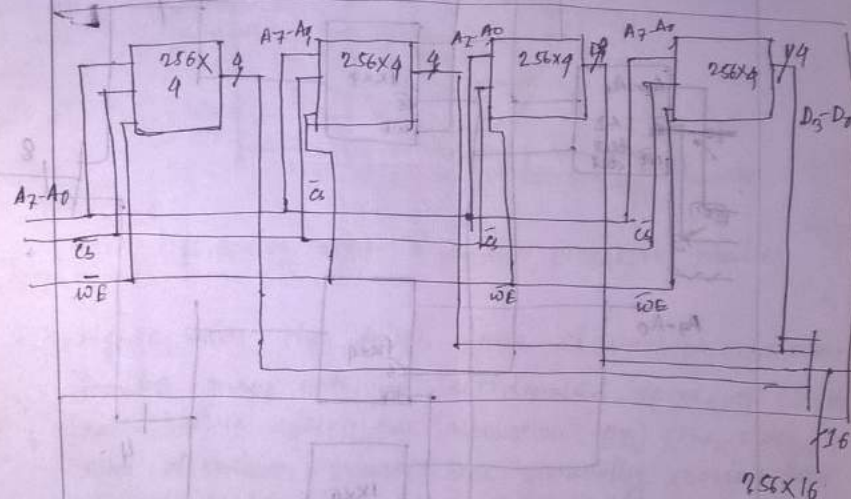
Output $1K \times 9$ $1K = 1024 = 2^{10}$ $A_0 - A_9$



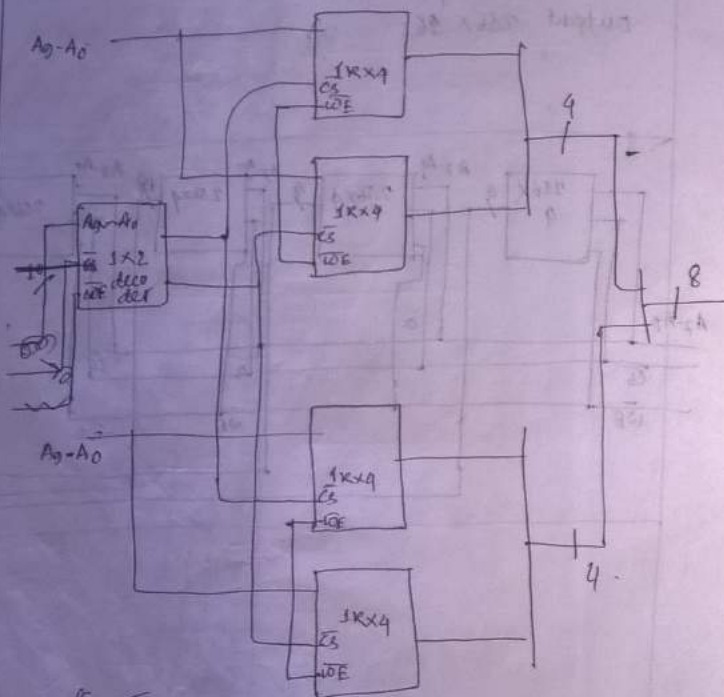
* Design 256×16 using 256×9 $1K \times 16$

input 256×9

output 256×16



* Design 2Kx8 using 1Kx4



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diagram

Design space of processors

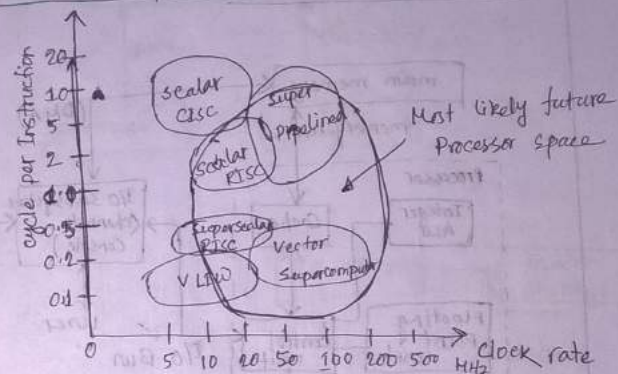


Fig: Design space of modern processor families.

Figure shows the design space of various processor families map onto a coordinated space of clock rate versus cycles per instruction (CPI). The clock rates of various processors are gradually moving from low to higher speeds towards the right of the design space. Processor manufacturers are trying to lower CPI. Based on these trends, the mapping of processor reflect the implementation during the past decade.